



BFHD: Bidirectional Feature Harmonization Decomposition for Heterogeneous Clinical Assessments

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Identify the largest feasible measurement subsets – without forcing all measurements to align.

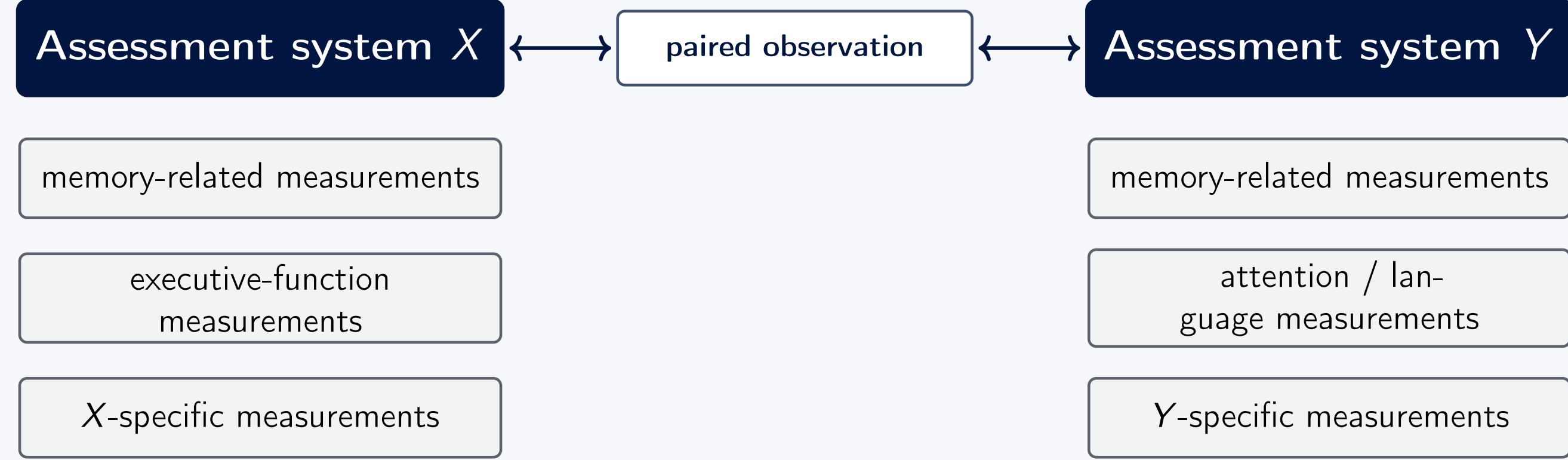


Code and supplementary material

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1. Clinical motivation

Different centres, cohorts, and time periods often use different but related clinical assessments. Their measurements partially overlap, while each system also contains information that is specific to its own protocol.



Key question

Which original measurements can be reliably translated across protocols, and how can we maximize harmonization coverage under an acceptable tolerance?

2. Existing methods and gap

Total-score linking

Aligns scores or low-dimensional traits, but is too coarse to identify recoverable measurements.

Predictive mapping

Learns $X \rightarrow Y$, or two separately parameterized directions, without intrinsic bidirectional coupling.

Latent alignment

Aligns latent representations across views, but does not explicitly decompose the original clinical measurements.

3. New problem: bidirectional recoverability

Paired systems and candidate subsets.

$$X \in \mathbb{R}^{d_X}, \quad Y \in \mathbb{R}^{d_Y}, \quad S_X \subseteq I_X, \quad S_Y \subseteq I_Y,$$

where $I_X = \{1, \dots, d_X\}$ and $I_Y = \{1, \dots, d_Y\}$.

Bidirectionally coupled model class.

$$f_\Phi: X \rightarrow Y, \quad g_\Phi: Y \rightarrow X.$$

The two directions are derived from a shared parameterization or structural constraint. Optimizing one direction therefore induces corresponding updates in the other, rather than allowing two unrelated mappings to drift.

ε -recoverability. A measurement is recoverable when its population discrepancy falls below an application-level tolerance:

$$\mathbb{E}[d(X_i, g_\Phi(Y)_i)] < \varepsilon, \quad \mathbb{E}[d(Y_j, f_\Phi(X)_j)] < \varepsilon.$$

A triple (S_X, S_Y, Φ) is ε -feasible when these conditions hold for every $i \in S_X$ and $j \in S_Y$:

$$(S_X, S_Y, \Phi) \in \mathcal{F}_\varepsilon.$$

Interpretation of ε

ε specifies clinically acceptable agreement. Smaller values impose stricter harmonization.

4. Maximize coverage subject to feasibility



BFHD optimization problem

$$\max_{\substack{S_X \subseteq I_X, S_Y \subseteq I_Y, \\ \Phi \in \mathcal{M}}} w|S_X| + (1-w)|S_Y| \quad \text{subject to } (S_X, S_Y, \Phi) \in \mathcal{F}_\varepsilon.$$

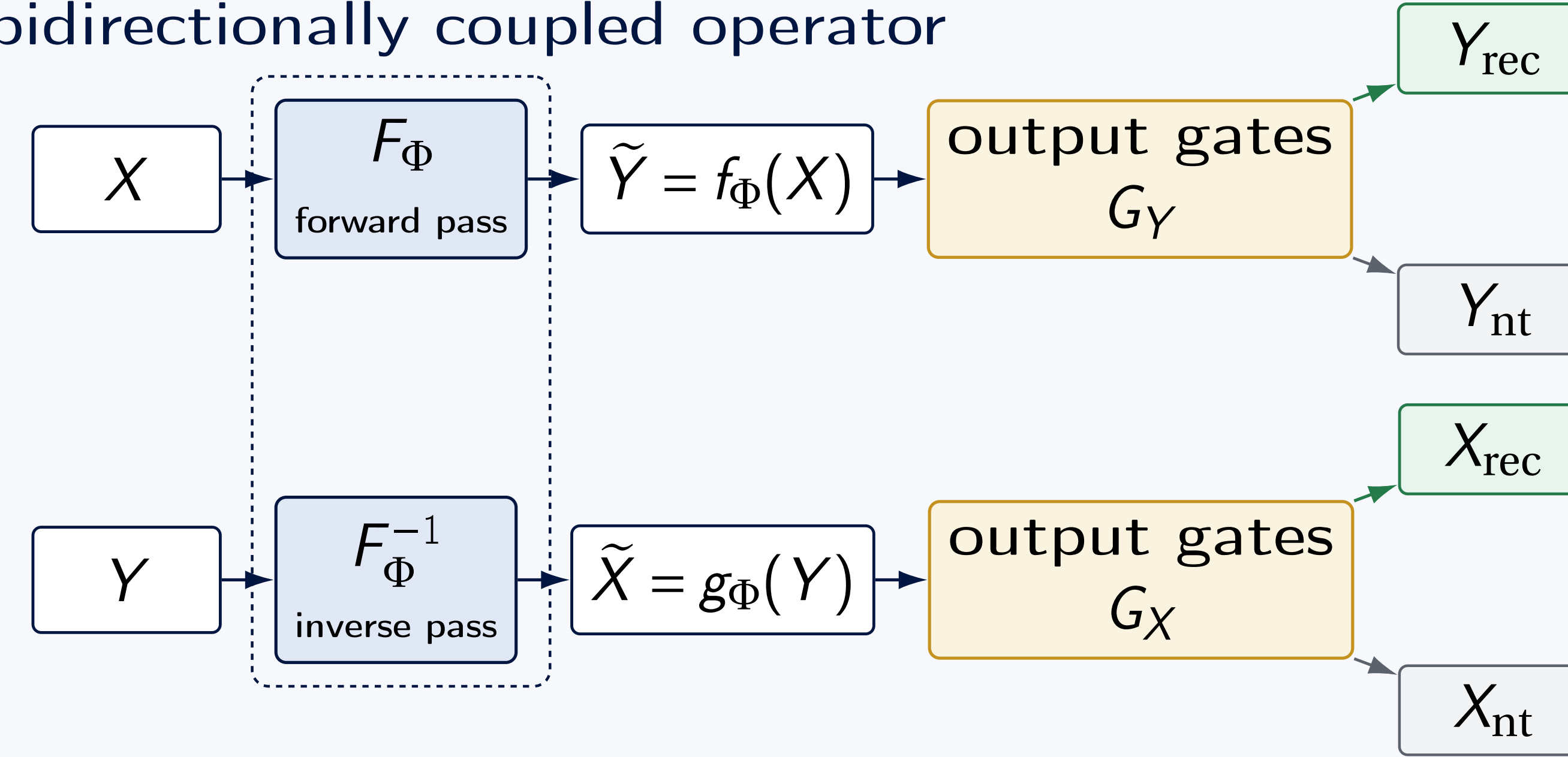
Unless otherwise stated, $w = 0.5$. An optimal solution yields

$$X_{\text{rec}} = X_{S_X^*}, \quad X_{\text{nt}} = X_{I_X \setminus S_X^*}, \quad Y_{\text{rec}} = Y_{S_Y^*}, \quad Y_{\text{nt}} = Y_{I_Y \setminus S_Y^*}.$$

Maximize measurement coverage subject to feature-wise feasibility.

5. BFHD: a practical relaxation of the formulation

one bidirectionally coupled operator



Strict coupling. The main implementation uses a NICE-style invertible backbone, so $X \rightarrow Y$ and $Y \rightarrow X$ are the forward and inverse passes of one operator.

Output-level decomposition. The model always receives the full opposite assessment; gates decide which predicted measurements must be reconstructed.

For inference:

$$\hat{Y}_j = G_{Y,j} \tilde{Y}_j, \quad \hat{X}_i = G_{X,i} \tilde{X}_i.$$

Important modelling point

Invertibility is used only to enforce strict parameter sharing. It is *not* an assumption that the clinical assessment systems themselves are invertible; after output gating, the overall mapping is intentionally non-invertible.

6. Feasibility-driven objective

$$\mathcal{L}_{X \rightarrow Y} = \sum_{i=1}^{d_X} G_{X,i} \text{ReLU}(\text{err}(X_i, \tilde{X}_i) - \varepsilon_{\text{train}}),$$

$$\mathcal{L}_{Y \rightarrow X} = \sum_{j=1}^{d_Y} G_{Y,j} \text{ReLU}(\text{err}(Y_j, \tilde{Y}_j) - \varepsilon_{\text{train}}),$$

$$\mathcal{L}_{\text{gate}} = -\lambda_{L_0} \left(\sum_{i=1}^{d_X} G_{X,i} + \sum_{j=1}^{d_Y} G_{Y,j} \right),$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{X \rightarrow Y} + \mathcal{L}_{Y \rightarrow X} + \mathcal{L}_{\text{gate}}.$$

Hinge feasibility: penalizes only violations of $\varepsilon_{\text{train}}$, creating pressure to improve reconstruction or reduce the corresponding gate.

Gate reward: favours inclusion and keeps gates open, unless feasibility pressure reduces the corresponding gate.

Staged training: backbone warm-up $\rightarrow$ joint feasibility optimization $\rightarrow$ gate hardening and backbone fine-tuning.

7. Experimental design

Dataset	Paired assessment systems	Role
Synthetic	$n = 2000$, $d_X = d_Y = 20$; known recoverable / non-translatable indices	Ground-truth structure recovery
ADNI	EMBIC (13) $\leftrightarrow$ NeuroBAT (9), $n = 1,629$	Real-world harmonization and temporal deployment
ADHD	CAS (27) $\leftrightarrow$ SNAP-IV (18), $n = 336$	Real-world harmonization

Baselines: IRT, feature-wise random forest (FRF), XGBoost (XGB), cycle-consistent MLP (CMLP), and NICE. All experiments use 10 random seeds. Real-world results use the reference operating point $\varepsilon_{\text{cutoff}} = 0.6$, with full ε -sweeps reported.

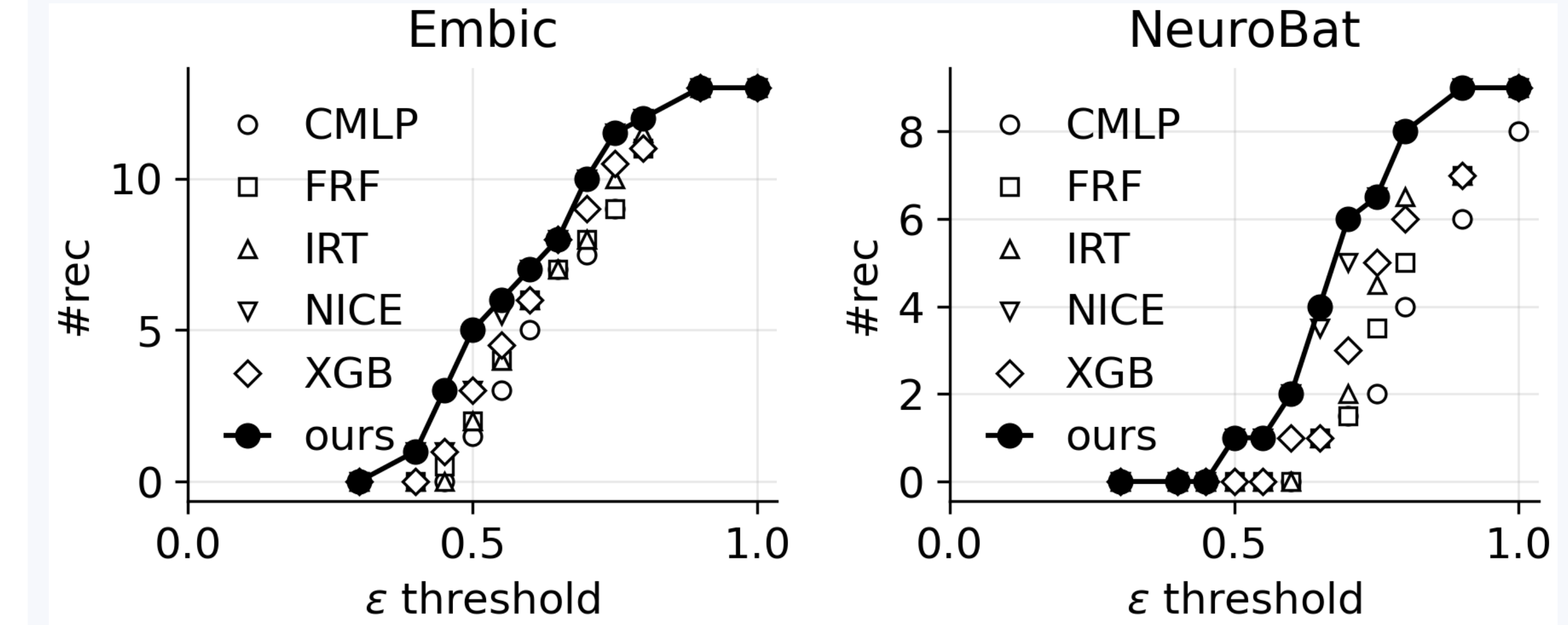
8. Results: recover the intended structure and expand the feasible frontier

Synthetic ground-truth recovery ($s_{\text{gen}} = 5$; mean $\pm$ std across 10 training seeds)

Method	Acc_X	Acc_Y	J_X	J_Y
CMLP	0.855 ± 0.015	0.810 ± 0.049	0.975	0.789
NICE	0.735 ± 0.039	0.725 ± 0.040	0.837	0.814
BFHD	0.880 ± 0.024	0.830 ± 0.024	0.933	0.924

BFHD gives the best overall index-level structure recovery while maintaining strong subset stability.

Real-world coverage–feasibility frontier



Across a broad range of ε , BFHD recovers more measurements than the baselines, particularly in the intermediate regime where only part of each assessment is mutually recoverable.

CAS $\leftrightarrow$ SNAP at

$\varepsilon_{\text{cutoff}} = 0.6$

#Rec_{A/B}: 11/9 for NICE and BFHD

$J_{A/B}$: 0.867/0.774 $\rightarrow$ 0.930/0.817

EMBIC $\leftrightarrow$ NeuroBAT at

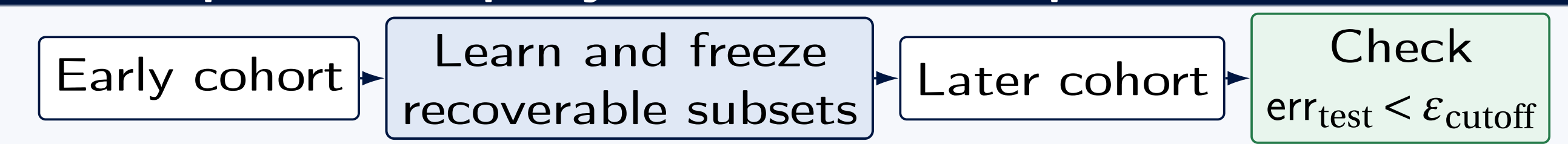
$\varepsilon_{\text{cutoff}} = 0.6$

#Rec_{A/B}: 7/2 for NICE and BFHD

$J_{A/B}$: 0.806/0.523 $\rightarrow$ 0.910/0.689

At the reference threshold, BFHD reaches maximal feasible coverage while improving subset stability over the ungated NICE backbone.

9. Temporal deployment under protocol drift



Method	#Ret _A	Ret% _A	#Ret _B	Ret% _B
IRT	3	42.9	0	N/A
XGB	6	85.7	0.5	62.5
FRF	5	56.0	0	0.0
CMLP	6	71.0	1	90.0
NICE	6	77.4	2	49.2
BFHD	7	100.0	3	100.0

BFHD retains 7 EMBIC and 3 NeuroBAT measurements, with 100% retention in both directions and no test-period re-selection.

10. Conclusion

- Coupled bidirectionality:** both directions are defined by one parameterization and constrain each other.
- Largest feasible subsets:** BFHD maximizes measurement coverage while requiring every selected measurement to satisfy the feasibility tolerance.
- Original-space decomposition:** recoverable and non-translatable measurements remain explicit in the original measurement spaces.